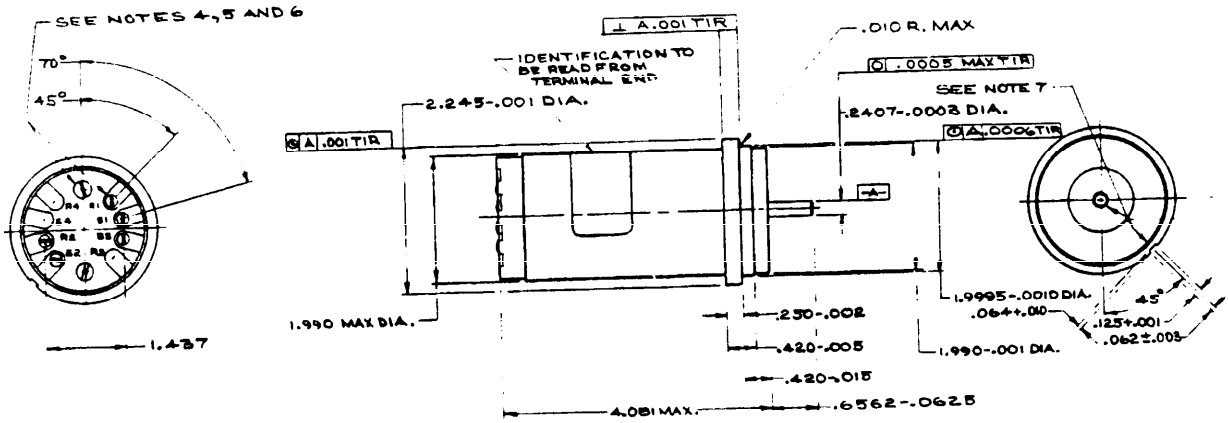
MIL-S-20708/142(MU)
31 July 1968

MILITARY SPECIFICATION SHEET

SYNCHRO CONTROL TRANSFORMER, TYPE 26V-23CT4a (10539690)

The complete requirements for procuring the synchro described herein shall consist of this document and the issue in effect of MIL-S-20708.



FSC 5990

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NOTES:

1. Dimensions are in inches.

2. Unless otherwise specified, tolerances on decimals are .XXX ± .005 and angles ± 1°.

3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.

4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Mounting screw positions optional. Terminal block designed for MS17182 terminal lug. (An approved 8 terminal block may be used provided all unused terminal positions are closed).

5. Minor variations of unit configuration are permitted for undimensioned design detail.

6. The following list of hardware shall be packaged in a ventilated envelope made of material having a PH value of 6-8 and compatible with the weight of the contents. The packaged hardware shall be placed in the unit synchro package as follows:

Machine screw	(MS 35275-225)	5 required
Lockwasher	(MS 35338-136)	5 required

7. Mark on shaft to align with slot or mark on housing within 10° when rotor is set at approx. synchro zero, and will bear no relation to the center lines shown on the drawing.

8. The synchro covered herein shall have 24 equally spaced slots to incorporate the stator windings.

9. The angular accuracy test stand employed for electrical error tests shall be capable of positioning the synchro rotor within ± 2 seconds of arc.

10. The null shift test shall be performed at -55° and 125°C.

TABLE I

REQUIREMENT	VALUE	UNIT	TOLERANCE
Frequency	400	Hz	± 1%
Primary Voltage	20.4	volts (rms)	± 1.2
Primary Voltage, Test	20.4	volts (rms)	± 1.2
Primary Current	1.2	milliamps	nominal
Primary Power	0.002	watts	nominal

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TABLE I (Continued)

REQUIREMENT	VALUE	UNIT	TOLERANCE
Impedance		ohms	R±15%+jx±10%
Zso	1400+j17500		
Zro	735+j7475		
Zrs	660+j460		
D.C. Resistance		ohms	±15%
Rotor	400		
Stator	620		
Transformation Ratio	0.635		± 0.02
Phase Shift	2.0 (lead)	degrees	± 0.5
Electrical Error	20	seconds	maximum
Null Voltage		millivolts	maximum
Total	7		
Fundamental	5		
Friction Torque	0.2	oz.-in.	maximum
@ -55°C	0.4		
Radial Play	0.001	inches	maximum
End Play	0.002	inches	maximum
Temperature Rise	20	degrees (C)	maximum
Null Shift (See Note 10)	4	sec./deg.(C)	maximum

Custodian:
Army - MU

Review activities:
Army - MU

User activities:
Army - MU

Preparing activity:
Army - MU

Project No. 5990-A012

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